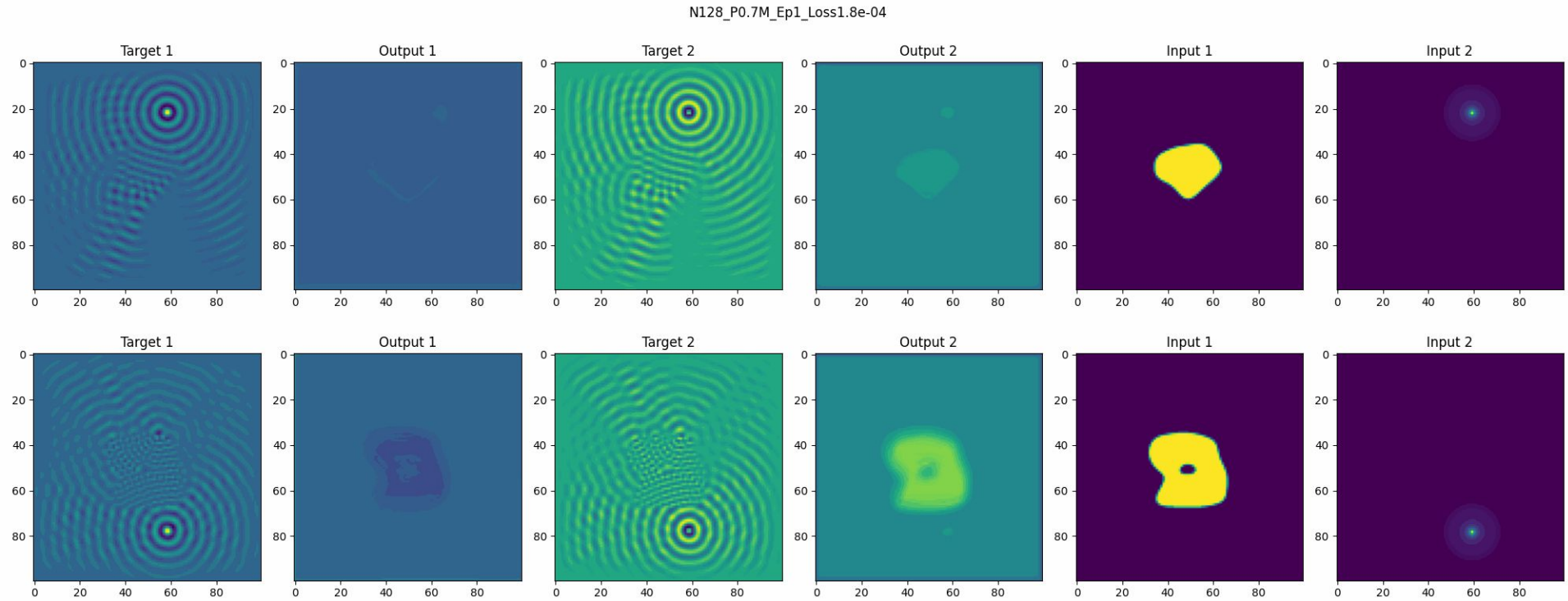
Progress Report

Data Stage


N128_P0.7M_Ep1_Loss1.8e-04
Target 1
Output 1
Target 2
Output 2
Input 1
Input 2
Target 1
Output 1
Target 2
Output 2
Input 1
Input 2

Model

Why Kronos for me?

- Project aligns with my physics and ML skills
  - Industry opportunities are unrelated to physics/math

- Rapid feedback system & Learning from the best.

Viska (Sijia) Wei 韦思嘉

EDU & Internships:

Berkeley （2012-2017) - Physics
UChicago（2017-2019) - Particle Pheno
JHU (2019-2025) - Physics Informed AI
JP Morgan, ByteDance - LLM Researcher

LLM Researcher
ByteDance · Internship
May 2024 - Aug 2024 · 4 mos
San Jose, California, United States · On-site

1. Built a next-generation model-as-a-service platform, supporting hundreds of LLM-based applications.
2. Developed and maintained the platform, covering offline training/finetuning, online inference, model management, and resource orchestration.
3. Implemented Automatic Prompt Optimization strategies and integrated Argilla for enhanced data processing.
4. Researched 60+ prompting and agent techniques (e.g., AutoPrompt, RL-GPT, OIRL, OPRO) and authored technical documents with 200+ team views.

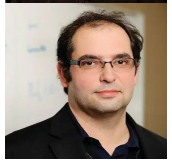
AI/ML Researcher
JPMorgan Chase & Co. · Internship
Jun 2023 - Aug 2023 · 3 mos
New York City Metropolitan Area · On-site

Major contribution to (1) Model design, coding, training and testing. (2) Workshop and tutorial creation for conceptual (NLP/LLM, ML/AI, InstructGPT) and practical (AWS, Pytorch) education.

1. Developed scalable sanction screening models based on large language models (RoBERTa, T5, Flan, Falcon-instruct, and GPT) that out-perform human operators, saving $20M in labor cost per year.
2. Utilizing efficient fine-tune techniques (quantization) and adapters (LoRA) to overcome GPU barriers, results in affordable and efficient development cycle that saves $2.6M per tasks per year.
3. Lectured on NLP/AI. Tech support on AWS, git, conda, and LLM libraries (trl, transformers).

Thesis Advisors & Research:

Pheno: Prof. Carlos Wagner
Phy: Prof. Alex Szalay
CS: Prof. Vova Braverman
Math: Dr. Sergey Kushnarev



Bounding the charm Yukawa coupling
NM Coyle, CEM Wagner, V Wei
Physical Review D 100 (7), 073013

A UMAP-based clustering method for multi-scale damage analysis of laminates
D Yang, V Wei, Z Jin, Z Yang, X Chen
Applied Mathematical Modelling 111, 78-93

Symmetric Norm Estimation and Regression on Sliding Windows
V Braverman, V Wei, S Zhou
COCOON 2021

Sketch and Scale Geo-distributed tSNE and UMAP
V Wei, N Ivkin, V Braverman, AS Szalay
2020 IEEE International Conference on Big Data (Big Data), 996-1003

Multiscale modelling for fatigue crack propagation of notched laminates using the UMAP clustering algorithm
D Yang, M Ma, V Wei, J Li, J Zhou, X Song, Z Guan, X Chen
Thin-Walled Structures 199, 111819

Tech Background - Graduate Level Courses:

PHYSICS	212	NONEQUIL STAT PHYS	4.0	4.0	A-
PHYSICS	221B	QUANTUM MECHANICS	5.0	5.0	A+
PHYSICS	232B	QUANTUM FLD THRY II	4.0	4.0	A+
PHYSICS	221A	QUANTUM MECHANICS	5.0	5.0	A
PHYSICS	226	PPP	4.0	4.0	A
PHYSICS	232A	QUANTUM FLD THRY I	4.0	4.0	A
PHYSICS	231	GEN RELATIVITY	4.0	4.0	A+
PHYSICS	233A	STND MOD AND BYND I	4.0	4.0	A
			Att	Earned	Gr Units
Term GPA	4.000	Term Totals	14.0	14.0	11.0

Course		Description	Attempted	Earned	Grade
MATH	31700	Topology/Geometry-1	100	100	A
PHYS	26400	Spacetime and Black Holes	100	100	A
PHYS	46000	Gravitational Waves	100	100	P
Winter 2018					
Course		Description	Attempted	Earned	Grade
MATH	31800	Topology/Geometry-II	100	100	A-
MATH	39701	Low-Dimensional Topology	100	100	A
PHYS	36400	General Relativity	100	100	A
Spring 2018					
Course		Description	Attempted	Earned	Grade
PHYS	44500	Quantum Field Theory-3	100	100	A
PSMS	39800	MS Research	100	100	P
PSMS	39900	MS Thesis/Project Research	100	100	A
Autumn 2018					
Course		Description	Attempted	Earned	Grade
PHYS	33000	Math Methods Of Physics-1	100	100	A
PSMS	39800	MS Research	200	200	P

EN.553.620 (01)	- Introduction to Probability [+]	A+
EN.553.626 (01)	- Introduction to Stochastic Processes [+]	A+
EN.553.627 (01)	- Stochastic Processes and Applications to Finance [+]	A+
EN.553.628 (01)	- Stochastic Processes and Applications to Finance II [+]	A+
EN.553.636 (04)	- Introduction to Data Science [+]	A+
EN.601.664 (01)	- Artificial Intelligence [+]	A+
EN.601.668 (01)	- Machine Translation [+]	A+
EN.601.633 (02)	- Intro Algorithms [+]	A+
AS.171.801 (14)	- Independent Research- Graduates [+]	A+
AS.171.749 (01)	- Machine Learning for Scientists [+]	A
EN.601.682 (01)	- Machine Learning: Deep Learning [+]	A

Course		Description	Attempted	Earned	Grade
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PHYS	26400	Spacetime and Black Holes	100	100	A
PHYS	46000	Gravitational Waves	100	100	P

Winter 2018

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MATH	39701	Low-Dimensional Topology	100	100	A
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Spring 2018

Course		Description	Attempted	Earned	Grade
PHYS	44500	Quantum Field Theory-3	100	100	A
PSMS	39800	MS Research	100	100	P
PSMS	39900	MS Thesis/Project Research	100	100	A

Autumn 2018

Course		Description	Attempted	Earned	Grade
PHYS	33000	Math Methods Of Physics-1	100	100	A
PSMS	39800	MS Research	200	200	P

53.620 (01)	- Introduction to Probability [+]	A+
53.626 (01)	- Introduction to Stochastic Processes [+]	A+
53.627 (01)	- Stochastic Processes and Applications to Finance [+]	A+
53.628 (01)	- Stochastic Processes and Applications to Finance II [+]	A+
53.636 (04)	- Introduction to Data Science [+]	A+
01.664 (01)	- Artificial Intelligence [+]	A+
01.668 (01)	- Machine Translation [+]	A+
EN.601.633 (02)	- Intro Algorithms [+]	A+
AS.171.801 (14)	- Independent Research- Graduates [+]	A+
AS.171.749 (01)	- Machine Learning for Scientists [+]	A
EN.601.682 (01)	- Machine Learning: Deep Learning [+]	A

PHYSICS	212	NONEQUIL STAT PHYS	4.0	4.0	A-
PHYSICS	221B	QUANTUM MECHANICS	5.0	5.0	A+
PHYSICS	232B	QUANTUM FLD THRY II	4.0	4.0	A+
PHYSICS	221A	QUANTUM MECHANICS	5.0	5.0	A
PHYSICS	226	PPP	4.0	4.0	A
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